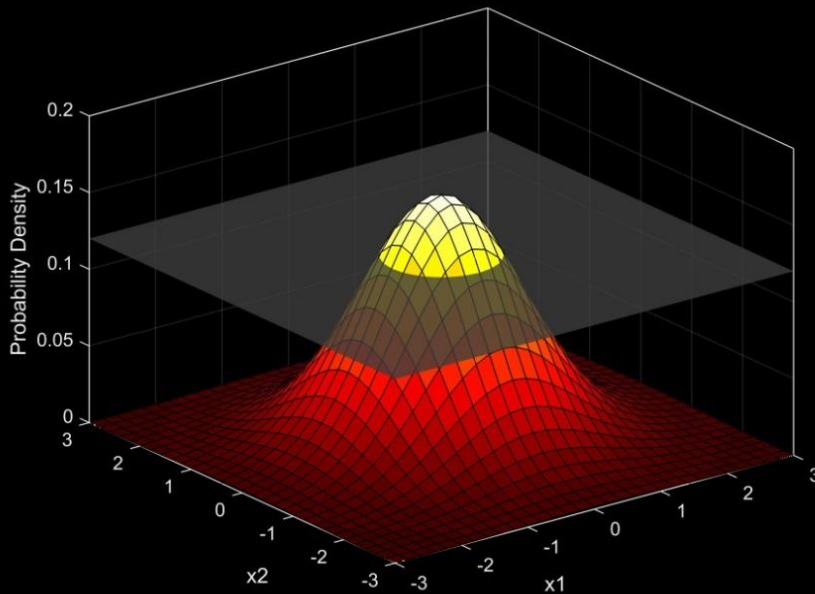


Bayesian Analysis in Ecology

WLF 5580, Spring 2027



$$[\theta|y] = \frac{[y|\theta][\theta]}{[y]}$$

Theory and practice of building and analyzing Bayesian models of ecological data

Learn:

- How Bayesian analysis yields probabilistic interpretations of parameters.
- What is MCMC, and how to implement.
- Selecting priors and diagnosing model convergence.
- Code with R and JAGS to simulate data and estimate parameters.
- Simple regression, hierarchical models, occupancy, mark-recapture, survival, N-mixture models, state-space models, and integrated models.
- Bayesian model selection techniques.
- How to make inferences and communicate technical science.

Prerequisites are STAT 431 or equivalent, proficiency in R, and familiarity with common probability density and mass functions. Consult instructor before enrolling.

Spring of odd years, 3 credits
Tu & Th 9:30 – 10:45

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